

DONGJIN SEO

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RESEARCH INTERESTS

My research is focused on the intersection of physics and artificial intelligence.

I apply physics to improve AI systems and use AI to advance our understanding and control of physical systems.

- *How can AI be used to design and discover new physical structures? (Inverse Design)*
- *How can AI learn the representation of physical systems? (Physical Inductive Bias)*
- *How can we embed physical systems into AI algorithms? (Physical Neural Networks)*

EDUCATION

[3] Yale University Ph.D. in Applied Physics (Academic Advisor: Prof. Logan G. Wright)	Aug 2024 - Connecticut, USA
[2] Korea Advanced Institute of Science and Technology (KAIST) M.S. in Electrical Engineering (Academic Advisor: Prof. Min Seok Jang) [thesis]	Feb 2019 - Feb 2021 Daejeon, South Korea
[1] Korea Advanced Institute of Science and Technology (KAIST) B.S. in Electrical Engineering - On Leave Aug 2014 - May 2016 for National Military Service	Feb 2011 - Feb 2019 Daejeon, South Korea

JOURNAL

[6] Physics-guided and fabrication-aware structural optimization using diffusion models D Seo [†] , S Um [†] , S Lee, JC Ye*, H Chung*. <i>ACS Photonics</i> [paper] [source code]	2026 <i>selected as the Supp Cover of 2026 Jan. Issue</i>
[5] ASOptimizer TM : optimizing antisense oligonucleotides through deep learning for IDO1 gene regulation G Hwang [†] , M Gwon [†] , <u>D Seo</u> , DH Kim, K Lee, E kim, M Kang*, J Ryu*. <i>Molecular Therapy Nucleic Acids</i> [paper]	2024
[4] Sample-efficient inverse design of freeform nanophotonic devices with physics-informed reinforcement learning C Park [†] , S Kim [†] , W Jeong [†] , J Park, <u>D Seo</u> , Y Kim, C Park, CY Park*, MS Jang*. <i>Nanophotonics</i> [paper]	2024
[3] Adjoint Method in Machine Learning: A Pathway to Efficient Inverse Design of Photonic Devices C Kang [†] , <u>D Seo</u> [†] , S V Boriskina, H Chung*. <i>Materials & Design</i> [paper]	2024
[2] Structural Optimization of a One-Dimensional Freeform Metagrating Deflector via Deep Reinforcement Learning D Seo [†] , DW Nam [†] , J Park, CY Park*, MS Jang*. <i>ACS Photonics</i> [paper] [source code] [press]	2022 <i>selected as the Front Cover of 2022 Feb. Issue</i>
[1] Inverse design of organic light-emitting diode structure based on deep neural networks S Kim, JM Shin, J Lee, C Park, S Lee, J Park, <u>D Seo</u> , S Park, CY Park, MS Jang*. <i>Nanophotonics</i> [paper]	2021

PREPRINTS

- [1] Wave Interpolation Neural Operator: Interpolated Prediction of Electric Fields Across Untrained Wavelengths
J Seo[†], C Kang[†], D Seo, H Chung.
[\[arXiv\]](#)

CONFERENCE

- [9] [poster] Physics-guided Optimization of Photonic Structures using Denoising Diffusion Probabilistic Models
D Seo†, S Um†, S Lee, J Ye, H Chung. *NeurIPS 2024 Workshop: Machine Learning and the Physical Sciences*
[\[extended abstract\]](#) Dec 2024
- [8] [poster] Wave Interpolation Neural Operator: Interpolated Prediction of Electric Fields Across Untrained Wavelengths
J Seo†, C Kang†, D Seo, H Chung. *NeurIPS 2024 Workshop: Data-driven and Differentiable Simulations*
[\[extended abstract\]](#) Dec 2024
- [7] [poster] Adjoint sensitivity analysis based photonic structure efficiency prediction and data augmentation
C Kang†, D Seo†, S V Boriskina, H Chung. *CLEO-PR 2024*
[\[proceeding\]](#) Aug 2024
- [6] [poster] High-Speed Multiwavelength Adjoint Optimization with Surrogate Solver
J Seo†, C Kang†, D Seo, H Chung. *CLEO-PR 2024*
[\[proceeding\]](#) Aug 2024
- [5] [poster] Physics-guided Diffusion Models for Inverse Design
D Seo†, S Um†, J Ye, H Chung. *CLEO-PR 2024*
[\[proceeding\]](#) Aug 2024
- [4] [poster] Contextualized and Aligned Audio-Text Fusion Models for Emotion Recognition
S Choi, Y Kwon, D Seo. *KCC 2023*
[\[proceeding\]](#) Dec 2023
- [3] [oral] Adjoint Method for Data Augmentation of Photonic Structures
D Seo, C Kang, H Chung. *Optica Imaging Congress*
[\[proceeding\]](#) Aug 2023
- [2] [oral] Deep reinforcement learning enables freeform structure optimization of 1D metagrating deflector
D Seo, DW Nam, J Park, CY Park, MS Jang. *SPIE Optical Engineering + Applications*
[\[abstract\]](#) Oct 2022
- [1] [poster] Realization of large scale graphene plasmonic resonator using epsilon-near-zero substrate
S Kim, S Baek, SY Min, H Ha, D Seo, J Kim, G Lee, B Min, MS Jang. *NANO KOREA 2022*
[\[abstract\]](#) Jul 2022

PATENT

- [10] Method and system for psychological test based on brain signal analysis
Korean Patent / Registration No. 10-2741867-0000 / Registration Date 2024.12.13
Inventors: D Seo, T Hwang.
- [9] Method and system for interactive psychological test
Korean Patent / Registration No. 10-2738489-0000 / Registration Date 2024.11.29
Inventors: S Choi, D Seo, T Hwang.
- [8] Device and method for placing classroom placements using student personality and grade data and machine learning technology
Korean Patent / Registration No. 10-2671422-0000 / Registration Date 2024.05.28
Inventors: S Choi, D Seo, T Hwang.
- [7] Method for optimizing classroom structure to achieve maximum learning efficiency utilizing policy-based reinforcement learning
Korean Patent / Registration No. 10-2671423-0000 / Registration Date 2024.05.28
Inventors: D Seo, T Hwang.

- [6] Devices, methods and programs for sampling a group of respondents based on artificial intelligence
Korean Patent / Registration No. 10-2663479-0000 / Registration Date 2024.04.30
Inventors: Y Kwon, S Choi, D Seo, T Hwang.
- [5] Method and System for Determining Psychological State based on Large Language Model
Korean Patent / Registration No. 10-2624653-0000 / Registration Date 2024.01.09
Inventors: S Choi, D Seo, T Hwang.
- [4] Server and Method for Generating Personality Test using Query Response Network based on Language Model
Korean Patent / Registration No. 10-2591769-0000 / Registration Date 2023.10.17
Inventors: Y Kwon, S Choi, D Seo, T Hwang.
- [3] Method for Sampling Process of Personality Test Using Question and Answer Network Representing Group of Respondents Based on BERT
Korean Patent / Registration No. 10-2583818-0000 / Registration Date 2023.09.22
Inventors: Y Kwon, S Choi, D Seo, T Hwang.
- [2] Method and System for Designing Optimal Sequence of RNA Therapeutics
Korean Patent / Registration No. 10-2546977-0000 / Registration Date 2023.06.20
Inventors: D Seo, M Kang, G Hwang, K Lee.
- [1] Method and System for Designing RNA Therapeutics
Korean Patent / Registration No. 10-2499895-0000 / Registration Date 2023.02.09
Inventors: D Seo, M Kang, G Hwang, K Lee.

HONORS AND AWARDS

- [7] Kwanjeong Scholarship [\[website\]](#) 2024 - 2030
- Korean scholarship to support doctoral program
- [6] 2nd Place of “2023 Corning AI Challenge” [\[website\]](#) Dec 2023
- [5] 6th Place of “AI Grand Challenge: Policy Assistance AI” Second Round [\[press\]](#) Dec 2023
- hosted by *the Ministry of Science and ICT of South Korea*
- Position: Team Leader
- Subject: Understanding and creating tables and figures, writing reports with a clear hierarchy using AI
- [4] 3rd Place of “AI Grand Challenge: Policy Assistance AI” [\[website\]](#) [\[press\]](#) Jul 2023
- hosted by *the Ministry of Science and ICT of South Korea*
- Position: Team Leader
- Subject: Developing an AI for the interpretation of governmental documents using NLP and CV techniques
- [3] 2022 Talent Award of Korea [\[website\]](#) [\[press\]](#) Dec 2022
- Award for talented people in South Korea
- bestowed by *the Deputy Prime Minister and Minister of Education of South Korea*
- [2] Best Paper Award (Honorable Mention) [\[website\]](#) Sep 2017
- bestowed by *the School of Humanities & Social Science, KAIST*
- [1] Exemplary Soldier Award May 2016
- bestowed by *Senior Superintendent of the Guard of Government Complex Daejeon*
(one person per platoon, Top 5%)

SERVICE

- [2] Reviewer at *NeurIPS ML4PS Workshop* 2024
- [1] Reviewer for *Nanophotonics* 2024

INVITED TALKS

- [3] Invited Talk at McMahon Lab, Cornell University, *Physics-guided and fabrication-aware inverse design using diffusion models* 2 Jul 2025
- [2] Invited Talk at KC ML2, *Recent research trends* 25 Jan 2024
- [1] Invited Talk at META Research Lab, MIT, *Structural optimization of one-dimensional freeform metagrating deflector via deep reinforcement learning* 9 Jun 2023

TEACHING

- [1] Teaching Assistant, *Statistical Methods with Applications in Science and Finance*, Yale University (Fall 2025)